724FM.2 ROSACEA TREATMENT PATHWAY

For Buckinghamshire use only

Rosacea is a chronic relapsing disease of the facial skin. It may improve with treatment, but intermittent relapses may occur. It is characterised by symptoms of facial flushing and a spectrum of clinical signs, including erythema, telangiectasia, coarseness of skin, and an inflammatory papulopustular eruption resembling acne. Epidemiological data are scarce and controversial; reported prevalence ranges from 0.09% to 22%. The aim of treatment should be complete skin clearance, where possible.

Referral criteria

Refer **routinely** to Dermatology:

- Flushing, persistent erythema, telangiectasia, or phymatous rosacea that have not responded to treatments described in Table 2 or are causing psychological or social distress.
- Papulopustular rosacea that has not responded to at least 12 weeks of oral plus topical treatment.
- An uncertain diagnosis.

Refer to an ophthalmologist:

- **Urgently**, if keratitis or anterior uveitis is suspected (eye pain, blurred vision, sensitivity to light, red eye).
- **Routinely**, if ocular symptoms are severe or resistant to maximal treatment in primary care.

Refer to a plastic surgeon if there is:

- Prominent non-inflamed phymatous disease (e.g. rhinophyma). This is not NHS funded and will need individual funding request (IFR) approval. Laser ablation can also be offered in the private sector.

If patient does not meet referral criteria, treat in primary care

- Assess the predominant symptoms and rule out acne vulgaris.
- Be aware that patients often experience more than one subtype concurrently.
- The treatment choice should be based on rosacea subtype and level of severity.

Self-care advice

Provide all patients with a Rosacea Lifestyle Patient Information Leaflet (see Patient Resources below):

- Provide reassurance about the benign nature of rosacea and uncommon progression to severe disease (especially in women).
- Recommend frequent application of high-factor sunscreen (minimum SPF 30) to the face whenever exposure to sunlight is likely. Ultraviolet protection sunglasses may be helpful for people with features of ocular rosacea. Avoid the use of sunbeds.
- If flushing is problematic, advise avoidance of trigger factors. Possible triggers include extremes of weather (heat and cold winds), sunlight, strenuous exercise, stressful situations, spicy food, alcohol, and hot drinks. Some drugs can aggravate flushing (e.g. calcium-channel blockers). A diary may be helpful to identify stimuli and triggers that may exacerbate rosacea.
- If the skin is dry, advise the use of skin-care products as required (e.g. hypoallergenic and non-comedogenic emollient creams). Avoid use of abrasive products or topical corticosteroids on the face. The possible use of yellow- or green-tinted cosmetics to help camouflage skin erythema.

Patient Information resources

- NHS Choices Rosacea: https://www.nhs.uk/conditions/rosacea/
- British Association of Dermatologists Rosacea Patient Information updated Aug 2018 http://www.bad.org.uk/shared/get-file.ashx?id=229&itemtype=document
- The British Skin Foundation: https://www.britishskinfoundation.org.uk/

Table 1: Rosacea Subtypes

Erythematotelangiectatic rosacea	Flushing and persistent erythema of the central face; possible telangiectases; easily irritated facial skin. Burning and stinging may be reported. Oedema, roughness, or scaling may be present.	
Papulopustular rosacea	Persistent erythema with transient papules and/or pustules of the central face. Burning and stinging may be reported.	
Ocular rosacea	Watery or bloodshot appearance; foreign body sensation, burning or stinging, dryness, itching, light sensitivity, blurred vision, telangiectases of lid margins; lid and periocular erythema; blepharitis, recurrent conjunctivitis, styes (chalazion, hordeolum); episcleritis, iritis; decreased visual acuity due to corneal complications (keratitis may occur).	
Phymatous rosacea	Skin thickening, irregular surface nodularities, and enlargement. Patulous follicles and telangiectases may occur. Rhinophyma is most common, but other affected locations may include the chin, forehead, cheeks and ears.	

Table 2: Treatment of Rosacea

	Flushing, erythema, telangiectasia [9, 10]	Papulopustular rosacea (mild to moderate)	Papulopustular rosacea (moderate to severe) and phymatous rosacea	Ocular rosacea
1st line (green traffic light status)	• Continue lifestyle changes throughout treatment and • Propranolol PO 10 mg - 40 mg twice daily (unlicensed indication) **Persistent erythema:** Brimonidine 0.5% gel[1] OD on an 'as needed' basis, for temporary relief of symptoms to maximum of 1 g of gel in total weight, which corresponds to approximately five pea sized amounts per day. **Ensure that patient knows how to apply the gel.** Topical Brimonidine may reduce erythema within 30 minutes, reaching peak action at 3 to 6 hours, after which the effect diminishes and erythema returns to baseline.	Ivermectin 1% cream[3] OD for 8 to 12 weeks If clinical improvement, continue maintenance therapy as needed, ideally until skin is clear. If little/no clinical improvement, use combination treatment as per moderate/severe papulopustular rosacea (adjacent).	Ivermectin 1% cream[3] OD for 8 to 12 weeks **AND** Doxycycline modified release 40mg capsule[5] OD for 8 to 12 weeks If no improvement at 6 weeks, consider discontinuing (licensed for maximum 16 weeks but review before this time). If clinical improvement, continue combination treatment up to 12 to 16 weeks, then reassess need for ongoing oral antibiotic treatment and continue topical treatment. **Clinically inflamed phymatous disease[11]:** Doxycycline modified release 40 mg capsule[5] PO for 6 weeks (off licence)	• Lid hygiene [8] • Artificial tears [9] Review after 2 to 4 weeks of self-care If any associated skin symptoms are seen or suspected, also treat as guidelines state for either as '**Papulopustular rosacea (mild to moderate)** or '**Papulopustular rosacea (moderate to severe)**' (adjacent) If clinical improvement, continue management as needed. If little or no improvement, refer to ophthalmology. Consider referral to dermatology if skin symptoms prominent.
2nd line (green traffic light status, use if 1st line ineffective, contra-indicated or not tolerated)	• Clonidine PO 50 micrograms twice daily (unlicensed indication) • Camouflage cream[2]	If ivermectin is not available or inappropriate e.g. severe hepatic impairment, pregnant or breastfeeding women:	**Alternative TOPICAL options** (if ivermectin is not available or inappropriate e.g., severe hepatic impairment, pregnant or breastfeeding women)**:**	

		Azelaic acid 15% gel BD, review at 2 months and if no improvement, discontinue gel [3] **OR** Metronidazole 0.75%[4] topical cream or gel BD for 6 to 9 weeks.	Azelaic acid 15% topical gel[4] BD, review at 2 months and if no improvement, discontinue gel **OR** Metronidazole 0.75% topical cream or gel BD for 3 to 4 months **Alternative <u>ORAL</u> options:** Oxytetracycline 500 mg[6] PO BD for 6 to 12 weeks, repeated intermittently **OR** Tetracycline 500 mg[6] PO BD for 6 to 12 weeks and are repeated intermittently **OR** Lymecycline 408 mg PO OD for 6 to 12 weeks, repeated intermittently (unlicensed indication) **<u>Alternative in impaired renal function:</u>** Doxycycline 100 mg po OD (off licence) **<u>Alternative for pregnant or breastfeeding women or when tetracyclines are contraindicated:</u>** Erythromycin 500 mg PO BD If little or no improvement, refer to Dermatology.	

Notes	1. When persistent, constant, severe erythema is the predominant symptom and there is no prominent telangiectasia. Brimonidine gel can also be used with metronidazole or azelaic acid gel in papulopustular rosacea: Apply metronidazole or azelaic acid after brimonidine gel has dried. 2. The charity 'Changing Faces' provides recommendations about camouflage creams. This is not an NHS service. Details of Changing Faces can be found on their website. Camouflage creams are not funded by the NHS and patients pay for them. 3. Ivermectin cream: Use one application daily for up to 4 months (review before this). If no improvement after 6 weeks consider discontinuing. The treatment course may be repeated as needed. 4. Topical metronidazole or azelaic acid can be used intermittently or continuously to control symptoms. Metronidazole cream is more cost effective than metronidazole gel and is the preparation of choice. It is also the preferred choice for sensitive skin. Metronidazole gel should be reserved for patients who have intolerance to metronidazole cream. Azelaic acid may be more effective in those who do not have sensitive skin but may cause transient stinging. 5. There is moderate-to-high-certainty evidence for doxycycline 40 mg modified release capsules for reducing papules/pustules. 6. There is low-certainty evidence for tetracycline. Only Oxytetracycline is licensed for rosacea. Lymecycline (off-license) or Doxycycline 100 mg (off-license) may be used as they are more effective, better tolerated and are taken as a single daily dose. 7. Eyelid hygiene: Apply a warm compress (clean flannel under hot water, wrung out and applied to lid for 1 minute) followed by cleansing with an over the counter lid cleaning solution or wipes. Refer to an ophthalmologist where necessary (see referral criteria). 8. Apply artificial tears (hypromellose 0.5% or carbomer 980 0.2%) liberally throughout the day into the eye. Chloramphenicol 1% eye ointment may be applied after eyelid hygiene at night. 9. Flushing, erythema/telangiectasia is a chronic condition which may burn itself out over several years. Treatment is aimed at temporary symptomatic relief. PRN treatment is an option. Daily dosing is also an option. 10. Telangiectasia is treated with laser therapy. This is not an NHS funded treatment (may only be treated for private patients). 11. Nasal debulking by laser ablation or surgical intervention are alternatives in people with significant rhinophyma. They are not NHS funded. Laser ablation is available privately, and surgical intervention will need prior IFR approval. **Other less well evidenced alternatives usually prescribed in secondary care should be noted (see Table 3 for doses):** • Other topical treatments, e.g., benzoyl peroxide, other topical antibiotics, tacrolimus, or retinoids (e.g., isotretinoin). • Other oral antibiotics, such as clarithromycin, azithromycin (useful if erythromycin is poorly tolerated), or minocycline. • The combined oral contraceptive (COC) pill (if a hormonal cause is suspected in a woman). • Oral isotretinoin. • Oral spironolactone (only if the rosacea has a suspected/confirmed acneiform element).

Table 3: Prescribing Information

Product Name	Dose

Brimonidine tartrate 3 mg/g gel	• Apply once daily when facial erythema is present. Brimonidine gel may only be needed on days when people feel that their appearance is particularly important. Apply thinly, divide dose over forehead, chin, nose and cheeks.
	• The maximum daily dose is 1 g of gel in total weight, which corresponds to approximately five pea-sized amounts.
	• Its use should be reviewed annually, including potential withdrawal.
	• Telangiectasia may be accentuated as general redness is reduced.
	• Some patients may have exacerbation or rebound symptoms of rosacea. It is important to initiate treatment with a small amount of gel (less than the maximum daily dose) for at least a week and increase the dose gradually, based on tolerability and treatment response.
	• Topical brimonidine may reduce erythema within 30 minutes, reaching peak action at 3 to 6 hours, after which the effect diminishes and erythema returns to baseline.
	• Patients should be counselled on the importance of not exceeding the maximum daily dose, and advised to stop treatment and seek medical advice if symptoms worsen during treatment
	• Brimonidine (Mirvaso) gel is subject to two MHRA warnings:
	  ○ MHRA June 17: Risk of systemic cardiovascular effects: Counsel to avoid application to irritated or damaged skin, including after laser therapy in order to reduce the risk of systemic absorption leading to bradycardia, hypotension and dizziness.
	  ○ MHRA Nov 16: Risk of exacerbation of rosacea: Counsel patients to initiate brimonidine with less than the max dose for at least 1 week, then increase gradually, based on tolerability and response. They should also be advised of the importance of not exceeding the maximum daily dose, stopping treatment and seeking medical advice if symptoms worsen during treatment.

Ivermectin 1% cream	One application a day for up to 4 months. Ivermectin should be applied daily over the treatment course for 8 - 12 weeks If clinical improvement, continue maintenance therapy as needed, ideally until skin is clear. If little/no clinical improvement, use combination treatment as per moderate/severe papulopustular rosacea. No dosage adjustment is necessary in renal impairment; caution in patients with severe hepatic impairment
Metronidazole 0.75% cream or gel	Apply a thin layer to the affected areas of the skin twice daily. Areas to be treated should be washed with a mild cleanser before application.
Azelaic acid 15% cream	Apply to the affected areas twice a day and rub in gently. Approximately 0.5 g = 2.5 cm (1 inch) of gel is sufficient for the entire facial area. Improvement becomes apparent after 4 weeks of treatment.

Product Name	Dose
Doxycycline	Doxycycline modified release 40 mg capsule[5] OD for 8 - 12 weeks If no improvement at 6 weeks, consider discontinuing (licensed for maximum 16 weeks but review before this time). If clinical improvement, continue combination treatment up to 12 to 16 weeks, then reassess need for ongoing oral antibiotic treatment and continue topical treatment.
Oxytetracycline	500 mg PO twice daily on an empty stomach
Tetracycline	500 mg PO twice daily
Lymecycline (unlicensed)	408 mg PO once daily
Doxycycline (unlicensed)	100 mg PO once daily (alternative for impaired renal function)
Erythromycin	500 mg PO twice daily
Artificial tears	As required taking patient preference into consideration.
Propranolol (unlicensed)	10 to 40 mg PO twice daily
Clonidine (unlicensed)	50 micrograms PO twice daily
Spironolactone^	25 mg PO daily increased gradually to a maximum of 100 mg PO twice daily
Isotretinoin capsules (unlicensed) (hospital only – red)	500 microgram to 1 mg/kg PO daily for 20 weeks. Alternatively, 10 to 20 mg PO daily
Benzoyl peroxide 5% gel	Apply once daily, preferably after washing with soap and water
Benzoyl peroxide/clindamycin 5%/1% (Duac Once Daily®)	Apply once daily, preferably after washing with soap and water
Tacrolimus 0.03% or 0.1% ointment (amber initiation)	Apply sparingly once daily
Isotretinoin 0.05% gel	Apply once daily
Azithromycin	500 mg PO once daily for two weeks (useful if erythromycin is poorly tolerated)
Clarithromycin	250 to 500 mg PO twice daily
Minocycline m/r tablets (amber initiation)	100 mg PO once daily
Combined oral contraceptive (COC) pill (ethinyloestradiol 30 micrograms + norethisterone or levonorgestrel)	One tablet daily

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This guideline was adapted from the Dorset CCG Rosacea Primary Care Treatment Pathway November 2015. It has subsequently been updated following the publication of the SCAN Rosacea guideline 2021.

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See also:
South Central Antimicrobials Network (SCAN) Acne Vulgaris guideline last updated December 2021

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